

Interactive Peer Evaluation for Calibrated Trust in AI Advisers

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1. Research Problem and Central Idea

When several AI systems endorse the same answer, a user may reasonably expect more support than one system can provide. Yet those systems may repeat the same mistake. Consider one AI adviser that reports a shipment has arrived and three evaluators that endorse its answer. All cite summaries derived from one outdated notice; a revised log records a delay. The three endorsements all rest on the same outdated information. Asking which dated source supports each judgment could expose the problem. We call the system proposing an answer the *adviser* and the evaluators the *panel*. Their *evaluation record* shows the user who endorsed the answer, what evidence was checked, and what remains uncertain.

This project asks: *How do people weigh agreement among AI evaluators, discount shared sources, and adjust reliance after errors?* Our central hypothesis is that showing shared sources and the evidence from follow-up checks will help people reject incorrect advice, while preserving their willingness to use correct advice. We will compare this with showing a reliability estimate alone. Aims 1 and 2 will assess shared sources and develop follow-up questions. Aim 3 will test how people use the resulting evidence when deciding whether to rely on advice. The project addresses the Trust and Influence portfolio’s interests in drivers of trust, dynamic trust assessment, and human-machine joint performance. We will study how people develop trust in an AI adviser, maintain it, and adjust it after errors.

We distinguish three concepts. **Reliability** is the measured quality of advice on a specified task. **Trust** is a person’s expectation that the adviser will help, measured through reported beliefs. **Reliance** is the decision to use its advice. Trust can guide reliance without guaranteeing that reliance is appropriate [15]. Following work on appropriate reliance [19], we will measure beneficial acceptance of correct advice and resistance to incorrect advice separately. Higher reported trust or more frequent agreement with AI will not by itself establish improvement.

Prior work and open questions. Our PRD method aggregates peer models’ pairwise judgments through iterative ranking and studies discussion between evaluators [17]. IQA-EVAL generates and assesses multi-turn question-answering interactions, with persona-based evaluation agents validated against human ratings [16]. These methods give us starting points for combining imperfect judgments and testing an adviser through follow-up questions. Peer scores still need to be calibrated against correct answers, and simulated users cannot establish how people will respond.

Related research documents correlated model errors and the limits of evaluation panels [12, 13]. Human-AI studies show that explanations and interaction design affect reliance, with persistent errors even in cooperative question answering [2, 11]. Behavioral research finds an illusion of consensus even when people can identify shared sources [26]; other studies show sensitivity to shared evidence and clearer source-independence cues [4, 24]. With labeled source relations and an always-correct simulated adviser, Ueno et al. found greater reliance on independent than shared-source support [22]. Study 1 tests whether

this distinction holds when evaluators can be wrong, measuring harmful and beneficial adoption separately. We will examine whether people see the panel’s evidence as independent support, and how follow-up checks and later errors change their reliance.

2. Research Aims and Technical Approach

Aim 1: Estimate what peer agreement actually tells us

We will extend PRD from relative model ranking to assessing particular advice, building on work with panels of LLM judges [23]. Evaluators will first judge candidate answers independently, with identities hidden and answer order randomized, and record supporting passages and unresolved objections. We will retain these initial judgments before any discussion so we can distinguish agreement reached during discussion from agreement present at the start.

Classical models estimate observer error rates [5], jointly infer expertise and item difficulty [25], and discount copied sources in truth discovery [9]. Building on these foundations, we will fit a regularized model of reviewer skill, task difficulty, and shared source effects on development cases judged independently by human reviewers. A starting method will adjust peer weights using these estimates and group endorsements that rely on the same source. This separates an evaluator’s demonstrated judging ability from its rank as an answer generator. Source overlap is observable evidence of dependence; different model names or prompts will not be treated as proof of independence. Comparisons will include observer-error and expertise/difficulty models without source terms, copied-source weighting, unmodified PRD, majority vote, the development-selected best judge, and supervised aggregation given the same labels. We will test how accounting for evaluators’ shared sources changes both the panel’s assessment and the user’s reliance.

The output will combine an estimated probability that the advice is correct and supported by the supplied documents with its evidence and unresolved concerns. Calibration will use data separate from both method development and final testing. In the shipment example, three references to one notice will be distinguished from support drawn from separately checked records. We will deliberately duplicate sources and measure model errors to test when this distinction improves assessment. Without suitable reference judgments, agreement can support a relative comparison but cannot establish absolute correctness. Tests on new topics will assess where calibration fails and additional human checking is needed. The research question is how much independent checking is needed before a panel’s agreement becomes a useful reliability signal.

Aim 2: Ask questions that reveal weaknesses in the advice

We will extend IQA-EVAL to generate short exchanges that test the advice. Prior LM cross-examination detects factual errors through inconsistencies [3]; we will focus on where evidence comes from and whether it is current. We will measure error detection, correction, and harm separately before testing

their effects on human reliance. Candidate questions will request a precise supporting passage, challenge an apparent contradiction, or ask whether a newer record changes the answer. In the shipment example, the useful question concerns the date and source of the arrival claim. A request to explain the same answer more fluently may add no evidence.

The system will select up to three follow-up questions based on how often each type exposes errors on similar development cases and how much interaction it requires. Randomized trials of a finite set of question types will estimate which checks work for each failure type; fixed scripts and random selection provide baselines. The system will not see the reference answers during evaluation. We will score the original advice and any revision separately: detecting an error, correcting it, and persuading another evaluator are different outcomes.

We hypothesize that questions seeking new, relevant evidence will expose shared mistakes more effectively than requests for additional agreement. Comparisons will include independent peer judgments, ordinary peer discussion, extra samples from one model, and targeted questions with the same available documents and matched computational budgets. The evaluation record will state what was checked, which answer changed, and what remains unresolved. We will also measure damage from inappropriate challenges, including correct advice changed to an incorrect answer. Unjustified agreement and answer-flipping after challenges are documented risks [14, 20]. IQA-EVAL agents will help develop interactions; the human studies in Aim 3 will test how these interactions affect trust and reliance.

Aim 3: Explain how evaluation evidence changes human reliance

We will conduct two preregistered online studies using short document-based decisions with independently established answers. Following the judge-advisor paradigm [1], participants will first give their own answer and confidence, then review AI advice and an evaluation record before deciding again. This lets us measure whether participants adopt advice when they initially disagree with it. A correct final answer alone would not tell us that.

Study 1: AI consensus as a social cue. We will vary shared versus separately sourced support and randomly assign participants to three displays: an endorsement count, a calibrated reliability estimate, and the same estimate accompanied by checked sources and unresolved objections. Each participant will see one display format but encounter different source arrangements, correct and incorrect advice, and different task types. Advice content, correctness balance, panel size, and allocated reading time will be matched across display conditions. One comparison will hold the advice, endorsements, and numerical estimate fixed, changing only whether the display reveals that endorsements reuse one source. Comprehension questions will check whether participants notice the shared source. This isolates the behavioral effect of disclosure from improvements in the advice itself. Tasks will range from looking up one record to combining several reports, allowing us to test whether the effect depends on how easily an answer can be checked.

The main comparison will test whether adding source records helps people reject incorrect advice more when the sources are shared than when they are separate. To count as an improvement, the records must reduce adoption of incorrect advice in shared-source cases without reducing adoption of correct advice beyond a small, prespecified margin relative to the estimate-only display. Competing predictions are that the illusion persists despite comprehension [26], or that richer records make wrong advice more persuasive, as observed with erroneous LLM explanations [21]. Including both correct and incorrect advice lets us test this risk separately from any benefit.

Study 2: Trust repair after errors. People may avoid an algorithm after seeing it make an error [8], motivating established work on trust repair in human-autonomy interaction [6, 18]. Across repeated decisions, participants will receive feedback about outcomes. After a disclosed failure, we will randomly assign a corrected answer alone or the same correction plus the evidence from the checks that led to it. Subsequent cases will include both recurrence of the failure type and unrelated reliable advice. We will test whether the evidence helps participants distinguish recurring problems from advice they can still use. Explanations of why an aid errs can increase reliance even when trust is unwarranted [10]; showing the evidence might therefore increase reliance even when the same failure recurs. Comparing recurring failures with unrelated, reliable advice will distinguish these predictions. We initially plan for 200–300 participants per study. Power simulations using pilot data will determine the final sample sizes, accounting for the main comparisons and Study 1’s non-inferiority criterion.

Building on longitudinal trust-calibration theory [7], a sequential behavioral model will predict reliance from prior outcomes, self-confidence, estimated advice reliability, and the support participants perceive as independent. We will compare models that count endorsements as separate votes with models that discount endorsements from a common source. Their predictions will be compared on held-out participants and tasks against models based only on agreement counts or past accuracy. The key test is whether shared sources and evidence from follow-up checks explain reliance better than these simpler models. Brief trust ratings and forecasts of adviser accuracy will measure beliefs separately from choices. Randomly assigning displays lets us test their causal effects. Associations between reported beliefs and choices alone will not establish why those effects occur.

3. Evaluation, Feasibility, and Expected Outcomes

Computational evaluation will start with small sets of documents for scheduling and resolving conflicting reports, including outdated records, duplicated sources, and unresolved conflicts. Human reviewers will establish answers and evidence support without seeing evaluator identities or conditions. Ambiguous cases will be retained as unresolved rather than assigned artificial ground truth. Held-out documents, topics, and evaluator combinations will test transfer. We will report calibration error, error-detection precision and recall, harmful revisions, abstention, and total cost, including all model calls and checks.

Reliability comparisons will match the fraction of cases receiving an assessment. We will also report how calibration improves as more human-labeled cases become available.

Human evaluation will report beneficial adoption when the initial human answer is wrong and AI advice is correct, and harmful adoption when the human is right and AI advice is wrong. Cases where both are wrong will be analyzed separately. Final accuracy, time, workload, and reported trust will complement these measures. Analyses will account for repeated observations by participant and task, report uncertainty intervals, and preregister primary contrasts. We will report controlled error cases separately from tests using the error rate observed in practice. Participant studies will undergo institutional review and use informed consent. If better automated assessments do not help people make better decisions, that will show a limit of the way we present the evidence.

Team and work plan. PI Xinya Du will lead the work with graduate researchers at UT Dallas. The proposed three-year budget of \$450,000, including indirect costs, supports personnel, computation, annotation, participant compensation, and dissemination. IQA-EVAL’s validation against human ratings provides experience connecting automated evaluations with human judgments. Before collecting confirmatory data, the PI will seek advice on the study protocols and power analysis. **Year 1** will establish the reference cases, dependence measurements, and calibrated peer assessments. **Year 2** will develop follow-up checks and conduct Study 1. **Year 3** will conduct Study 2 and test transfer of the computational and behavioral models. Our PeerEval prototype provides a starting point for peer evaluation. We will develop the interfaces needed for the human studies.

Potential Air Force and Space Force impact. Applications could include AI-assisted reconciliation of logistics updates, maintenance records, and mission-status reports. Several evaluators may appear to provide independent support while relying on the same original report. This research would identify when to display an endorsement count, disclose shared sources, or request an additional check. It would also provide measures for testing whether those interventions reduce harmful reliance while preserving useful advice after a visible failure. Applying the methods in these settings would require further testing with the relevant users and records.

The project will produce models of how people respond to agreement among evaluators, methods for interactive peer evaluation, and reusable tasks and study protocols. We aim to identify when automated evaluation helps people judge advice and when they need more independent evidence.

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